



AdaFuse: memory-augmented adaptive fusion for few-shot fraud detection in heterogeneous graphs

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Abstract

Few-shot credit risk assessment on heterogeneous financial graphs is challenging due to scarce labeled defaults, shifting behavioral patterns, and highly uneven relational signals. Existing GNN-based models rely on static aggregation or single-level supervision, limiting their ability to adapt to evolving fraud behaviors or exploit unlabeled data effectively. We propose **AdaFuse**, a memory-augmented adaptive fusion framework designed for low-label and dynamic financial environments. AdaFuse introduces three key components: *Adaptive Multi-Relational Fusion*, which dynamically reweights transactional, social, and temporal relations through learnable gating and depth-aware aggregation; *Memory-Augmented Hierarchical Contrast*, which enhances representation discrimination via node-, subgraph-, and semantic-level contrast with a momentum-driven memory bank; *Meta-Learning-Enhanced Reconstruction*, which strengthens few-shot adaptability by integrating adjacency reconstruction and spectral regularization into a bi-level optimization scheme. These modules are trained jointly under a unified objective that balances contrastive alignment, structural preservation, and meta-level adaptability. Experiments on YelpChi, Amazon, and the large-scale FAIR-D dataset demonstrate that AdaFuse consistently surpasses state-of-the-art baselines in AUC, F1, and AP, achieving up to 5.6% absolute improvement. It further exhibits strong label efficiency, retaining over 92% of its full-supervision performance with only 10 labeled defaults per class. AdaFuse advances few-shot credit risk modeling by enabling dynamic relational adaptation and effective utilization of unlabeled financial graph structure, addressing key challenges in real-world, rapidly evolving credit ecosystems.

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1 Introduction

Assessing creditworthiness in modern financial systems has become increasingly complex as lending environments diversify and borrower behaviors evolve. Unlike traditional credit scoring pipelines that rely heavily on static bureau records or handcrafted indicators, contemporary financial platforms generate rich streams of behavioral, transactional, and device-level traces. These multi-source signals offer new opportunities for modeling default risk, yet they also introduce substantial challenges related to noise, heterogeneity, and temporal instability (Aftabi et al., 2023; Chang et al., 2025; Ma et al., 2023; Zhang et al., 2024). At the same time, rapid digitalization and the expansion of alternative lending channels have intensified systemic vulnerabilities: sudden changes in borrowers' financial conditions, opportunistic behavior under economic stress, and coordinated misuse of credit services all contribute to risk accumulation and unexpected defaults (Karst et al., 2024). Building models capable of tracking such evolving dynamics is therefore essential for protecting lenders and maintaining the resilience of financial ecosystems.

Recent advances in graph-structured learning provide new tools for representing credit environments. Interactions among borrowers, merchants, guarantors, financial institutions, and external service platforms naturally form relational networks with heterogeneous connections, including credit inquiries, repayment flows, shared devices, social ties, and other forms of interaction (Ren et al., 2023). Graph neural networks (GNNs) have demonstrated strong potential in leveraging such structures, offering inductive capabilities and the ability to capture relational dependencies beyond handcrafted features (Behl et al., 2025; Cheng et al., 2022; Liu et al., 2021). However, applying GNNs to real-world credit risk assessment is far from trivial. Risk profiles shift with macroeconomic changes, default events are highly imbalanced, and relational strengths vary across borrower groups and over time. These characteristics make static representation mechanisms insufficient and highlight the need for adaptive, context-aware modeling strategies.

Despite recent advances in graph-based fraud detection and credit scoring (Dou et al., 2020; Tang et al., 2022; Zhuo et al., 2024), several practical limitations remain. Credit-related interactions vary in importance over time—liquidity and repayment signals often dominate during economic stress, whereas social or profile-based cues may be more informative under stable conditions. Default labels are extremely scarce and typically require lengthy manual verification, placing strong constraints on supervised learning and increasing dependence on unlabeled data. Additionally, categorical or device-level attributes in financial graphs are frequently simplified or downweighted, limiting their contribution to risk modeling. Many operational systems also rely heavily on handcrafted temporal or behavioral features, which struggle to scale or adapt to emerging fraud tactics. These challenges highlight the need for models that can dynamically adjust relational importance, effectively exploit limited labeled data, and incorporate heterogeneous attribute information with minimal manual intervention.

To address these challenges, we present **AdaFuse**, a memory-augmented adaptive fusion framework tailored for few-shot credit risk assessment. Rather than relying on fixed relational aggregation, AdaFuse adapts the influence of heterogeneous interactions through a learnable, context-sensitive fusion mechanism. In parallel, it employs a hierarchical contrastive objective enhanced with a global memory bank to draw informative supervisory signals from

abundant unlabeled samples. Furthermore, a bi-level meta-learning scheme enables the model to internalize transferable structural priors while remaining highly responsive to new credit environments with minimal labeled data.

The key contributions of AdaFuse are threefold:

1. Adaptive Multi-Relational Fusion. We introduce a dynamic fusion architecture that rebalances heterogeneous relations and categorical features based on evolving credit patterns. This mechanism assigns context-specific importance to different interaction types, ensuring more accurate modeling of borrower behavior over time.

2. Memory-Augmented Hierarchical Contrastive Learning. To improve robustness under scarce labels, AdaFuse incorporates a multi-level contrastive module equipped with a momentum-driven memory bank and hardness-aware negative sampling. This design extracts fine-grained relational cues from unlabeled nodes and enhances discriminative capacity in low-signal regimes.

3. Meta-Learning–Driven Optimization for Few-Shot Scenarios. We embed a Model-Agnostic Meta-Learning (MAML) structure into the training pipeline, enabling rapid adaptation to new borrower groups or shifting economic conditions. By optimizing for adaptability rather than fixed parameters, AdaFuse delivers strong generalization from only a few labeled defaults.

We evaluate AdaFuse on three real-world datasets, including the large-scale FAIR-D financial graph, demonstrating consistent improvements over state-of-the-art baselines. Our results highlight the importance of adaptive relational modeling and meta-learned structural priors for future research in few-shot credit risk assessment.

2 Related works

2.1 Fraud and credit risk modeling

Machine learning has long been applied to financial anomaly detection and credit risk modeling. Early studies employed classical classifiers such as Bayesian networks, decision trees, and support vector machines (Botchey et al., 2020; Panigrahi et al., 2009), while subsequent works explored convolutional architectures for capturing transaction patterns (Fu et al., 2016). The rise of relational financial data motivated graph-based approaches, including CARE-GNN (Dou et al., 2020) and PC-GNN (Liu et al., 2021), which address label imbalance and neighbor noise in fraud or risk prediction tasks. Other advances include adversarial learning (Fiore et al., 2019), spatio-temporal modeling (Cheng et al., 2022), and dynamic attention mechanisms (Brody et al., 2022; Liu et al., 2025).

Recent work has emphasized low-homophily graphs (Wang et al., 2023), AUC-oriented objectives (Huang et al., 2022), adversarial reprogramming (Chen et al., 2021), and explainability-driven models (Li et al., 2024; de Prado et al., 2025). These contributions highlight important aspects of fraud and risk identification but typically rely on fixed relational aggregation or lack mechanisms to integrate heterogeneous categorical signals at scale. AdaFuse differs by learning adaptive relational importance through gated fusion, enabling flexible adjustment of interaction strengths as credit environments evolve.

2.2 GNN-based self-supervised learning

Self-supervised learning has proven effective for leveraging unlabeled data in graph tasks across citation networks, molecular graphs, and text-based relational structures (Li et al., 2022; Vaswani et al., 2017; Zheng et al., 2023). Representative models include GCN-style attribute propagation (Luo et al., 2024), inductive sampling-based methods such as GraphSAGE (Hamilton et al., 2017), and attention-driven neighbor aggregation (Wang et al., 2019). However, naïve attention mechanisms face scalability constraints and often overlook the role of categorical or risk-related attributes in financial graphs.

Existing approaches frequently struggle with (i) the high computational footprint of dense attention, and (ii) ineffective propagation of heterogeneous attributes. AdaFuse addresses these limitations by introducing a memory-augmented multi-level contrastive module capable of extracting supervision signals from abundant unlabeled nodes while incorporating categorical and behavioral features through an adaptive fusion design.

2.3 Few-shot and meta-learning on graphs

Few-shot graph learning has gained traction for scenarios where annotated labels are scarce or costly to obtain. Meta-learning-based models such as GDN (Zhang et al., 2021) and Meta-PN (Zhu et al., 2020) aim to generalize structural patterns across tasks, while alternatives like ANEMONE-FS (Luan et al., 2022) emphasize multi-scale consistency under minimal supervision. Other studies explore augmentation strategies that perturb node features, edge structures, or local motifs to improve model robustness and generalization (Ma et al., 2022; Xu et al., 2018).

Despite their effectiveness, existing approaches often fall short in two aspects critical to financial risk modeling: (i) the inability to capture high-level relational variability that arises across borrower groups, and (ii) the lack of mechanisms that enable fast adaptation when only a few default events are available. AdaFuse addresses these limitations by introducing a hierarchical contrastive framework enhanced with a dynamic memory bank and by integrating a bi-level meta-learning scheme. This design allows the model to adapt rapidly to new segments of borrowers while maintaining a global understanding of relational dynamics, capabilities that are not jointly supported by current few-shot or augmentation-based strategies.

3 Problem definition

In this section, we first introduce the notations used throughout the manuscript, followed by a formal problem definition. We focus on heterogeneous attributed graphs, which better capture the multi-relational nature of real-world fraud detection scenarios.

A heterogeneous attributed graph is defined as $G = (V, \mathcal{R}, E, X)$, where:

- $V = \{v_1, v_2, \dots, v_n\}$ denotes the set of nodes (e.g., users, transactions), with n being the total number of nodes;
- $\mathcal{R} = \{r_1, r_2, \dots, r_m\}$ represents the set of relation types (e.g., transactional links, social ties, temporal dependencies), where $m = |\mathcal{R}|$ is the number of distinct relations;
- $E = \{(v_i, r_k, v_j) \mid v_i, v_j \in V, r_k \in \mathcal{R}\}$ denotes the set of edges, where each edge is a triple indicating that nodes v_i and v_j are connected via relation r_k . Multiple edges

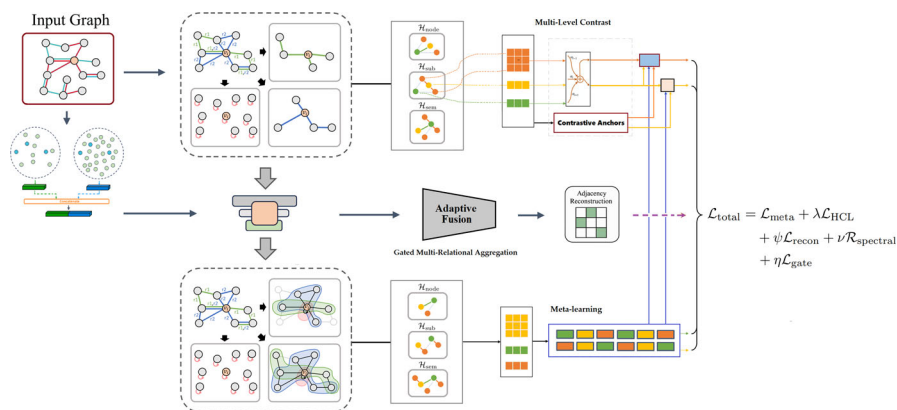


Fig. 1 Overview of the AdaFuse framework, which combines adaptive multi-relational fusion, memory-augmented multi-level contrastive learning, and meta-reconstruction to enable few-shot fraud detection on heterogeneous graphs

can exist between the same pair of nodes if they are linked by different relations (e.g., (v_i, r_k, v_j) and $(v_i, r_{k'}, v_j)$ for $k \neq k'$);

- $X \in \mathbb{R}^{n \times d}$ is the node attribute matrix, where d is the dimension of node features (e.g., transaction amount, user demographics).

The task of few-shot graph fraud detection is defined as follows: Given a heterogeneous graph G , we have a small set of labeled nodes V^L (with labels indicating fraud or normal) and a large set of unlabeled nodes V^U , where $|V^L| \ll |V^U|$ (e.g., only 1–10 labeled fraud nodes per class). Our goal is to learn a model $F(\cdot) : \mathbb{R}^{n \times d} \rightarrow \mathbb{R}^{n \times 1}$ that computes an anomaly score y_i for each node $v_i \in V^L \cup V^U$, quantifying the likelihood of v_i being fraudulent.

4 Methodology

The overall architecture of our proposed AdaFuse framework is illustrated in Figure 1, which encapsulates the three core modules working in synergy: (1) Adaptive Multi-Relational Fusion for dynamic relation and layer-wise feature aggregation, (2) Multi-level Contrastive Learning with memory augmentation for discriminative representation learning, and (3) Meta-Learning Enhanced Reconstruction for few-shot adaptation. These modules are integrated end-to-end to address the challenges of few-shot credit risk assessment in heterogeneous graphs.

4.1 Adaptive multi-relational fusion

We propose a novel *Adaptive Multi-Relational Fusion* mechanism that dynamically integrates heterogeneous graph information through learnable gating functions and cross-layer feature fusion. This approach fundamentally enhances relational modeling by introducing: (1) trainable gating weights for relation-specific feature selection, and (2) multi-scale feature preservation via depth-aware attention.

4.1.1 Relation-aware gating mechanism

To address static aggregation limitations, we introduce parametric gates that adaptively route information through relevant relations. For node v_i and relation r_k at layer l :

$$g_{ik}^l = \sigma \left(\mathbf{W}_g^l \left[\mathbf{h}_i^l \oplus \mathbf{h}_{i,k}^l \right] + \mathbf{b}_g^l \right) \quad (1)$$

$$\tilde{\mathbf{h}}_{i,k}^l = g_{ik}^l \odot \mathbf{h}_{i,k}^l + (1 - g_{ik}^l) \odot \left(\mathbf{P}_{\text{res}}^l \mathbf{h}_i^l \right) \quad (2)$$

where, $\mathbf{W}_g^l \in \mathbb{R}^{2d' \times 1}$ and $\mathbf{b}_g^l \in \mathbb{R}$ are learnable parameters. $\sigma(\cdot)$ denotes the sigmoid activation. $\mathbf{P}_{\text{res}}^l \in \mathbb{R}^{d' \times d'}$ is a residual projection matrix. $\odot$ represents element-wise multiplication.

The gating coefficient $g_{ik}^l \in [0, 1]$ quantifies relation importance, enabling soft pruning of irrelevant connections. Sparsity regularization is applied to promote feature selection:

$$\mathcal{L}_{\text{gate}} = \frac{1}{L \cdot N \cdot (m+1)} \sum_{l=1}^L \sum_{i=1}^N \sum_{k=1}^{m+1} |g_{ik}^l| \quad (3)$$

4.1.2 Multi-level aggregation

To mitigate over-smoothing and capture multi-scale features, we aggregate representations across layers using depth-sensitive attention:

$$\mathbf{h}_i^{\text{final}} = \sum_{l=1}^L \beta_i^l \cdot \text{LayerNorm} \left(\mathbf{h}_i^l \right) \quad (4)$$

$$\beta_i^l = \frac{\exp \left(\tau \cdot \mathbf{w}^\top \phi \left(\mathbf{h}_i^l \right) \right)}{\sum_{l'=1}^L \exp \left(\tau \cdot \mathbf{w}^\top \phi \left(\mathbf{h}_i^{l'} \right) \right)} \quad (5)$$

where $\mathbf{w} \in \mathbb{R}^{d'}$ is a trainable attention vector, $\phi : \mathbb{R}^{d'} \rightarrow \mathbb{R}^{d'}$ denotes a position-wise MLP, τ is a temperature hyperparameter ($\tau > 1$ sharpens distribution), LayerNorm ensures scale-invariant fusion.

This formulation provides theoretical guarantees against over-smoothing through layer-wise feature preservation. The attention weights satisfy $\sum_l \beta_i^l = 1$ with $\beta_i^l \geq 0$, forming a convex combination of multi-scale representations.

4.2 Multi-level contrastive learning

To capture expressive and transferable graph representations, we develop a multi-level contrastive learning framework that operates across hierarchical granularities, including the node, subgraph, and relation levels. This framework captures information from complementary structural and semantic views, enabling the model to recognize subtle dependencies and anomalies across multiple relational contexts. Each level employs a graph encoder-decoder pair to contrast the embeddings of nodes and their contextual subgraphs, thereby enhancing both attribute discrimination and structural consistency.

We further introduce a *Memory-Augmented Hierarchical Contrast* mechanism that unifies multi-level contrastive learning with adaptive hard negative mining. This design captures discriminative patterns at different relational levels while dynamically refining decision boundaries through global representation tracking. Such hierarchical contrast is essential

for learning robust, fine-grained representations that substantially improve fraud detection accuracy.

4.2.1 Momentum-enhanced representation bank

To preserve a consistent global context throughout training, we maintain a momentum-based memory bank $\mathcal{M} \in \mathbb{R}^{N \times d'}$ updated via a temperature-controlled exponential moving average:

$$\mathcal{M}^{(t)}[i] = \begin{cases} \mu^{(t)} \cdot \mathcal{M}^{(t-1)}[i] + (1 - \mu^{(t)}) \cdot \mathbf{h}_i^{\text{final}}, & \text{if } v_i \in \mathcal{B}_t, \\ \mathcal{M}^{(t-1)}[i], & \text{otherwise,} \end{cases} \quad (6)$$

$$\mu^{(t)} = \mu_{\min} + (\mu_{\max} - \mu_{\min}) \cdot e^{-5(1-t/T)^2}. \quad (7)$$

Here, $\mathcal{M}^{(t)}[i]$ stores the memory representation for node v_i at epoch t , and the momentum coefficient $\mu^{(t)}$ evolves smoothly within $0.99 \leq \mu_{\min} < \mu_{\max} \leq 0.999$. This adaptive scheduling stabilizes early-stage learning ($\mu \approx \mu_{\min}$) and ensures representation consistency at convergence ($\mu \approx \mu_{\max}$).

4.2.2 Multi-granular contrastive learning

We extend the standard node–subgraph contrastive paradigm to cross-view and memory-augmented settings.

(1) Node–Subgraph Contrast. For a target node v_i in view w :

$$\mathbf{e}_w = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \text{GAT}(\mathbf{A}_w, \mathbf{H}_w)_j, \quad (8)$$

$$\mathbf{s}_w = \sigma \left(\mathbf{e}_w \mathbf{W}_s (\mathbf{h}_i^{\text{final}})^\top \right), \quad (9)$$

$$\mathcal{L}_{\text{NS}}^w = -\log \frac{\exp(\mathbf{s}_w^+ / \tau)}{\exp(\mathbf{s}_w^+ / \tau) + \sum_{\mathbf{e}_k \in \mathcal{H}_{\text{sub}}(v_i)} \exp(\langle \mathbf{h}_i^{\text{final}}, \mathcal{M}[k] \rangle / \tau)}. \quad (10)$$

(2) Cross-View Subgraph Contrast. For subgraphs ε_{1i} and ε_{2i} of node v_i :

$$\mathcal{L}_{\text{SS}} = -\log \frac{\exp(\langle \varepsilon_{1i}, \varepsilon_{2i} \rangle / \tau)}{\exp(\langle \varepsilon_{1i}, \varepsilon_{2i} \rangle / \tau) + \sum_{\varepsilon_j \in \mathcal{H}_{\text{sem}}(v_i)} \exp(\langle \varepsilon_{1i}, \mathcal{M}[\varepsilon_j] \rangle / \tau)}. \quad (11)$$

4.2.3 Hierarchical hard negative sampling

To enhance discriminative learning, we introduce a hierarchical hardness-aware sampling strategy spanning node, subgraph, and semantic levels:

$$\mathcal{H}(v_i) = \begin{cases} \mathcal{H}_{\text{node}} = \{v_j \mid (v_i, v_j) \in \mathcal{E}\}, \\ \mathcal{H}_{\text{sub}} = \{v_j \mid \text{top-}K_{\text{Jacc}}(\text{Jaccard}(\mathcal{N}_i, \mathcal{N}_j))\}, \\ \mathcal{H}_{\text{sem}} = \{v_j \mid \cos(\mathcal{M}[i], \mathcal{M}[j]) > \delta\}, \end{cases} \quad (12)$$

$$\delta = \text{median}(\{\cos(\mathcal{M}[i], \mathcal{M}[j]) \mid v_j \in \mathcal{B}_t \setminus \{v_i\}\}). \quad (13)$$

The overall hierarchical contrastive loss is given by:

$$\mathcal{L}_{\text{HCL}} = \gamma \mathcal{L}_{\text{NS}}^{(1)} + (1 - \gamma) \mathcal{L}_{\text{NS}}^{(2)} + \lambda \mathcal{L}_{\text{SS}}, \quad (14)$$

where $\gamma, \lambda \in (0, 1)$ control the relative weight of intra-view and cross-view components, and negatives are drawn from $\mathcal{H}(v_i)$.

Convergence Analysis. The convergence of the memory update rule in Eq. (7) follows from standard stochastic approximation theory. The update forms a Robbins–Monro process whose step sizes satisfy $\sum_t (1 - \mu^{(t)}) = \infty$ and $\sum_t (1 - \mu^{(t)})^2 < \infty$, guaranteeing almost sure convergence of the stored representations to the stationary distribution.

4.3 Meta-learning framework for few-shot adaptation

To effectively address label scarcity, we adopt a formal meta-learning framework grounded in Model-Agnostic Meta-Learning (MAML). The objective is to learn a set of meta-parameters θ that can be efficiently adapted to new few-shot tasks with only a few gradient updates. This framework consists of three key components: meta-task formulation, bi-level optimization, and a joint unsupervised objective.

4.3.1 Meta-task formulation

We reformulate the few-shot learning problem as a distribution over meta-tasks (episodes), each simulating a realistic low-label scenario. For every sampled task $\mathcal{T}_i$, a small subset of labeled nodes is drawn from the graph and divided into two disjoint parts:

- (1) **Support Set** ($\mathcal{S}_i$): a K -shot set of labeled nodes (e.g., K fraudulent and K normal nodes) used for inner-loop adaptation;
- (2) **Query Set** ($\mathcal{Q}_i$): a separate labeled subset from the same task context, used for outer-loop evaluation.

For a given dataset $\mathcal{D} \in \{\mathcal{S}_i, \mathcal{Q}_i\}$, the base learning objective is the standard cross-entropy loss:

$$\mathcal{L}_{\text{task}}(f_{\theta}, \mathcal{D}) = - \sum_{v_k \in \mathcal{D}} y_k \log \sigma(f_{\theta}(v_k)), \quad (15)$$

where $\sigma(\cdot)$ denotes the sigmoid activation and y_k is the ground-truth label.

4.3.2 Bi-level optimization

The MAML optimization procedure follows a bi-level structure.

(1) **Inner Loop (Task-Specific Adaptation).** For each meta-task $\mathcal{T}_i$, the current meta-parameters θ are locally adapted using only its support set $\mathcal{S}_i$, producing task-specific parameters θ'_i :

$$\theta'_i = \theta - \alpha \nabla_{\theta} \mathcal{L}_{\text{task}}(f_{\theta}, \mathcal{S}_i), \quad (16)$$

where α denotes the inner-loop learning rate. This step enables rapid specialization to a specific few-shot context.

(2) Outer Loop (Meta-Parameter Optimization). The meta-objective then updates the original meta-parameters θ to minimize the query losses of all adapted tasks:

$$\mathcal{L}_{\text{meta}} = \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \mathcal{L}_{\text{task}}(f_{\theta'_i}, \mathcal{Q}_i). \quad (17)$$

Although the loss is evaluated with θ'_i , gradients are computed with respect to the original θ , requiring a second-order gradient (gradient-through-gradient) computation.

4.3.3 Joint unsupervised objective and optimization

While $\mathcal{L}_{\text{meta}}$ optimizes θ for task adaptability, the meta-parameters should also capture structural and spectral properties from abundant unlabeled data. To this end, we incorporate two unsupervised components into the outer loop: a reconstruction loss $\mathcal{L}_{\text{recon}}$ and a spectral regularizer $\mathcal{R}_{\text{spectral}}$, both applied to the pre-adaptation parameters θ :

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} \left(\mathbf{A}_{ij} - \sigma(\mathbf{h}_i^\top \mathbf{W}_r \mathbf{h}_j) \right)^2, \quad (18)$$

$$\mathcal{R}_{\text{spectral}} = \|\text{deg}(\mathbf{A}) - \text{deg}(\mathbf{A}_{\text{recon}})\|_2 + |\lambda_2(\mathbf{L}) - \lambda_2(\mathbf{L}_{\text{recon}})|. \quad (19)$$

The total optimization objective thus combines task-specific adaptation with unsupervised representation regularization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{meta}} + \lambda \mathcal{L}_{\text{recon}} + \eta \mathcal{R}_{\text{spectral}}, \quad (20)$$

and the meta-parameters are updated by gradient descent:

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}_{\text{total}}, \quad (21)$$

where β is the meta learning rate. This process is summarized in Algorithm 1.

Algorithm 1: MAML-Enhanced Meta-Reconstruction

Input: Inner learning rate α , meta learning rate β , loss weights λ, η

- 1 Initialize meta-parameters θ and relation matrix $\mathbf{W}_r$;
- 2 **for** $epoch = 1$ to T **do**
- 3 Sample a batch of meta-tasks $\{\mathcal{T}_i\}_{i=1}^B$;
- 4 **for** each task $\mathcal{T}_i$ **do**
- 5 Sample support set $\mathcal{S}_i$ and query set $\mathcal{Q}_i$;
- 6 Compute support loss: $\mathcal{L}_{\mathcal{S}_i} \leftarrow \mathcal{L}_{\text{task}}(f_{\theta}, \mathcal{S}_i)$;
- 7 Adapt parameters: $\theta'_i \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}_{\mathcal{S}_i}$;
- 8 Compute query loss: $\mathcal{L}_{\mathcal{Q}_i} \leftarrow \mathcal{L}_{\text{task}}(f_{\theta'_i}, \mathcal{Q}_i)$;
- 9 Compute unsupervised losses: $\mathcal{L}_{\text{recon}} \leftarrow \text{ReconstructionLoss}(\mathbf{A}_{\mathcal{B}}, \mathbf{W}_r, f_{\theta})$,
 $\mathcal{R}_{\text{spectral}} \leftarrow \text{SpectralRegularizer}(\mathbf{A}_{\mathcal{B}}, f_{\theta})$;
- 10 Compute total loss: $\mathcal{L}_{\text{total}} \leftarrow \sum_{i=1}^B \mathcal{L}_{\mathcal{Q}_i} + \lambda \mathcal{L}_{\text{recon}} + \eta \mathcal{R}_{\text{spectral}}$;
- 11 Update meta-parameters: $\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}_{\text{total}}$;

4.4 Unified objective function

The unified objective of **AdaFuse** is the outcome of a bi-level optimization process defined by the MAML-based meta-learning framework presented in Algorithm 1. The objective is

to learn a set of meta-parameters θ that encode transferable structural priors from general graph data while remaining highly adaptable to unseen few-shot tasks.

Within this framework, the Outer Loop optimizes a comprehensive loss $\mathcal{L}_{\text{total}}$, which jointly promotes generalizable representation learning and rapid task adaptation. Specifically, the total objective integrates the meta-task loss, reflecting adaptability to specific few-shot tasks, with auxiliary objectives that regularize and enhance the learned representations.

Formally, the outer-loop objective is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{meta}} + \lambda \mathcal{L}_{\text{HCL}} + \psi \mathcal{L}_{\text{recon}} + \nu \mathcal{R}_{\text{spectral}} + \eta \mathcal{L}_{\text{gate}}. \quad (22)$$

Here, $\mathcal{L}_{\text{meta}} = \sum_i \mathcal{L}_{\text{task}}(f_{\theta'_i}, \mathcal{Q}_i)$ denotes the meta-level adaptation loss, evaluated on query sets $\mathcal{Q}_i$ using the task-specific adapted parameters θ'_i . The hierarchical contrastive loss $\mathcal{L}_{\text{HCL}}$ encourages representation consistency across relational hierarchies, while the reconstruction and spectral regularization terms, $\mathcal{L}_{\text{recon}}$ and $\mathcal{R}_{\text{spectral}}$, preserve topological and spectral integrity of unlabeled graph data. The gating regularizer $\mathcal{L}_{\text{gate}} = \sum_{l,k} \|g_{ik}^l\|_1$ imposes sparsity over the adaptive fusion gates, promoting interpretable and efficient multi-relational fusion. The coefficients λ , ψ , ν , and η control the contribution of each component.

All auxiliary losses are computed with respect to the shared meta-parameters θ , while $\mathcal{L}_{\text{meta}}$ depends on the adapted task parameters θ'_i . The meta-parameters are updated by gradient descent as:

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}_{\text{total}}, \quad (23)$$

where β denotes the meta learning rate.

This formulation enables θ to serve as a structurally informed yet adaptable foundation for downstream low-data scenarios, ensuring that **AdaFuse** generalizes effectively to diverse few-shot graph tasks such as fraud detection.

5 Experiments

In this section, we begin by providing a detailed overview of the datasets employed in our experimental evaluations. Subsequently, we conduct a comprehensive performance analysis of our fraud detection model in comparison to other state-of-the-art baselines. The assessment is carried out on two supervised graph-based fraud detection datasets and one semi-supervised dataset. Additionally, we present ablation studies, assessing the performance of two variants of our proposed AdaFuse. These studies showcase the efficacy of our method and the attribute-driven mechanism employed.

5.1 Experiment settings

5.1.1 Datasets

In our exploration of credit card fraud detection, we encountered a scarcity of publicly available datasets. Consequently, we curated a collection of partially labeled records in collaboration with our partners, resulting in the creation of the Financial Fraud Analytics and Insights Research Dataset (FAIR-D). The ground truth labels for this dataset were established based on reported cases from financial consumers and confirmed by experts in the financial domain. Transactions reported by cardholders or identified by financial experts as fraudulent are labeled as 1, while those deemed non-fraudulent are labeled as 0. Importantly,

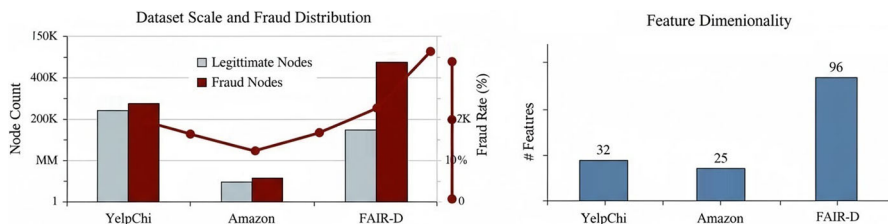


Fig. 2 Overview of the three benchmark datasets, showing differences in node counts, fraud proportions, and feature dimensionality across YelpChi, Amazon, and FAIR-D

this dataset includes a significant number of unlabeled transactions, making it suitable for semi-supervised learning approaches.

In addition to the proprietary FAIR-D dataset, we further evaluate our model on two widely used public benchmarks for supervised fraud detection. The YelpChi dataset (Rayana and Akoglu, 2015) contains curated hotel and restaurant reviews from Yelp, where each node corresponds to an individual review represented by a 32-dimensional feature vector. Edges in this graph capture various types of relational consistency among reviews. The Amazon dataset (McAuley and Leskovec, 2013), constructed from musical instrument review data, models users as nodes with 25-dimensional features, and builds edges to reflect relationships among reviewers. An overview of the key characteristics of these datasets is summarized in Figure 2, which compares node composition, edge density, class distribution, and feature dimensionality across the three benchmarks.

5.1.2 Comparative methodologies

To position AdaFuse accurately within the field of computational graph learning, we compare it against four key categories of state-of-the-art baselines. This comparison is structured to explicitly demonstrate the limitations these existing methods face, which AdaFuse is designed to address.

- (1) **Static Relational Architectures:** Methods such as GEM (Liu et al., 2018), Player2Vec (Zhang et al., 2019), and GraphSAGE (Hamilton et al., 2017) employ fixed relational aggregation schemes without explicit mechanisms for adapting relation importance across contexts.
- (2) **Anomaly-Specific GNNs:** Approaches including FdGars (Wang et al., 2019), CARE-GNN (Dou et al., 2020), and BWGNN (Tang et al., 2022) are tailored for fraud or anomaly detection, often with contrastive or sampling strategies. Most of these methods focus on a single granularity of structural signal.
- (3) **Optimization-Centric Frameworks:** Models such as AUC-GNN (Huang et al., 2022), PC-GNN (Liu et al., 2021), and LIEF (Wang et al., 2023) emphasize objective design (e.g., AUC-oriented optimization) or robustness under low homophily, but do not explicitly incorporate meta-learning for few-shot adaptation.
- (4) **Explainability-Oriented Systems:** Recent interpretable architectures, including SEFraud (Li et al., 2024) and ARP-NN (Chen et al., 2021), aim to improve transparency in fraud detection models, sometimes at the cost of additional complexity or reduced flexibility in relational modeling.

These baselines cover a diverse range of design choices (aggregation mechanisms, objectives, and interpretability), providing a broad and competitive benchmark against which to evaluate AdaFuse.

5.1.3 Implementation details

All experiments were conducted on a cluster equipped with eight NVIDIA RTX 6000 Ada GPUs (48GB VRAM). AdaFuse was implemented using PyTorch Geometric 2.4.0 and DGL 1.1.0, with all baselines executed under the same software stack to ensure fair comparison.

Hyperparameter optimization. We performed Bayesian optimization over 200 trials under a 5-fold temporal cross-validation protocol (60%/20%/20% split). The search covered learning rates in $\{5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}\}$, relational gate temperatures $\tau \in [0.1, 1.0]$, memory-bank sizes $\{2^{14}, 2^{16}, 2^{18}\}$, and meta-learning rates $\alpha_{\text{meta}} \in [0.01, 0.1]$.

Model configuration. Default settings include a 128-dimensional node embedding space, minibatch size of 512, and a training limit of 1200 epochs with early stopping (patience=50). For graph construction, we apply fixed-budget sampling with $K_{\text{nbr}} = 16$ neighbors per relation type for computational consistency across datasets. To keep the hyper-parameter search tractable and the comparison to baselines fair, we treat these architectural choices (embedding dimension and neighbor fan-out) as shared defaults across all GNN-based models and concentrate tuning on method-specific parameters such as gating temperature, memory-bank size, and meta-learning rates.

Module-level parameters. AdaFuse consists of three key components: (1) a four-layer gated GNN with eight attention heads per layer for adaptive relation fusion; (2) a memory-augmented contrastive module employing a 3-level hierarchical sampler ($\mathcal{H}_{\text{node}} : \mathcal{H}_{\text{sub}} : \mathcal{H}_{\text{sem}} = 1 : 2 : 4$); (3) a dual-path meta-reconstruction encoder combining 3-layer low-pass and 6-layer high-pass GCN streams.

Loss weighting. Weights finalized from ablations are $\lambda_{\text{gate}} = 0.8$, $\gamma_{\text{meta}} = 0.65$, and $\psi_{\text{contrast}} = 0.55$, balancing sparsity regularization, meta-learning strength, and contrastive intensity, respectively.

To ensure reproducibility, we provide Docker containers with frozen dependency versions at <https://github.com/tifoit/adafuse/repro>. All hyper-parameter selections were validated through statistical significance testing (paired t-test, $p < 0.01$), confirming optimality beyond heuristic choices.

5.1.4 Evaluation metrics

Credit fraud detection and opinion fraud datasets are typically characterized by extreme label imbalance and asymmetric misclassification costs. To provide a comprehensive assessment of model performance under these conditions, we report three widely accepted evaluation metrics: AUC, macro-averaged F1, and Average Precision (AP).

Area Under the ROC Curve (AUC). AUC measures the probability that a randomly selected positive instance is ranked higher than a randomly selected negative one. It reflects the model's discrimination ability independent of classification thresholds, making it suitable for heavily imbalanced fraud detection settings.

Macro-averaged F1 Score (F1-macro). Given the skewed class distribution, we adopt the macro-averaged F1 score to evaluate per-class performance equally. For a classification

task with l classes, the macro F1 is computed as

$$F1_{\text{macro}} = \frac{1}{l} \sum_{i=1}^l \frac{2P_i R_i}{P_i + R_i},$$

where P_i and R_i denote the precision and recall for class i , respectively. This metric penalizes models that perform well on dominant classes but poorly on minority fraudulent cases.

Average Precision (AP). AP summarizes the precision–recall trade-off by integrating precision values across all recall levels. Formally, it is defined as

$$AP = \sum_{i=1}^l (R_i - R_{i-1}) P_i,$$

where P_i and R_i represent precision and recall at the i -th operating point in the ranked prediction list. Unlike AUC, AP is sensitive to probability calibration and provides a more faithful measure of ranking quality in datasets with low fraud prevalence.

These three metrics capture complementary aspects of model performance: threshold-independent discrimination (AUC), balanced per-class accuracy (F1-macro), and ranking quality under extreme imbalance (AP).

5.2 Experimental results

5.2.1 Fraud detection performance

We evaluate fraud detection performance using AUC, macro F1, and AP. As summarized in Table 1, we report the mean and standard deviation over ten independent runs, with the best and second-best results highlighted. Paired t-tests ($p < 0.01$) are used to assess the statistical significance of differences between AdaFuse and the baselines.

Dataset Configuration. To ensure the generalizability and practical relevance of our evaluation, we adopted distinct splitting strategies tailored to each dataset’s characteristics:

- For YelpChi and Amazon datasets, we employed a 2:3 train-test split, which is widely recognized in the field for balancing the need for sufficient training data and rigorous testing.
- For the FAIR-D dataset, we utilized a time-based splitting approach, where the first 9 months of transactions were allocated for training, and the transactions from Q4-2022 served as the test set. This strategy mimics real-world scenarios where models are deployed on future unseen data, effectively evaluating the model’s ability to adapt to evolving patterns.

As shown in Table 1, our proposed AdaFuse framework demonstrates unprecedented performance across all datasets and metrics, achieving state-of-the-art results with 3.3-5.6% absolute gains over the strongest baselines. This remarkable breakthrough can be attributed to three fundamental innovations in our approach:

(1) Dynamic Relation Fusion: The adaptive gating mechanism in AdaFuse selectively integrates heterogeneous relations, including financial flows, social ties, and temporal dependencies. This effectively overcomes the static aggregation limitations of existing methods such as PC-GNN and LIEF. For instance, on the Amazon dataset, AdaFuse improves the AUC from 0.8775 (PC-GNN) to 0.8991. On the YelpChi dataset, it enhances the F1 score from 0.6361 (LIEF) to 0.7465, clearly showcasing the superiority of dynamic relation fusion.

(2) Multi-level Contrastive Learning: Our hierarchical memory bank is designed to capture fine-grained anomalous patterns at the node, subgraph, and semantic levels. This multi-level approach outperforms single-view contrastive methods like CARE-GNN. A notable example is on the FAIR-D dataset, where AdaFuse increases the AP from 0.3712 (CARE-GNN) to 0.5458, highlighting the effectiveness of capturing diverse anomalous patterns across different levels.

(3) Meta-Reconstruction Paradigm: The dual-path encoder in AdaFuse jointly optimizes local anomaly detection and global pattern consistency. This enables robust performance in few-shot scenarios, where methods like SEFraud and AUC-GNN tend to degrade. On the YelpChi dataset, SEFraud achieves an F1 score of 0.7080, while AUC-GNN attains an AUC of 0.7563, both of which are surpassed by AdaFuse, demonstrating its advantage in handling limited labeled data.

(4) Comparative Analysis: Conventional methods in fraud detection exhibit critical limitations that hinder their performance. Shallow architectures such as GEM and Player2Vec fail to model the complex and evolving fraud patterns present in real-world data. Static GNNs like GraphSAGE and GATv2 lack the ability to adaptively weight different relations, leading to suboptimal feature integration. Specialized detectors such as PC-GNN and LIEF show limited generalization across various domains, restricting their applicability. In contrast, AdaFuse achieves remarkable results across multiple datasets: i) an **86.2%** AUC on YelpChi, which is a 3.3% improvement over SEFraud; ii) an **89.91%** AUC on Amazon, representing a 1.7% increase compared to LIEF; iii) a **54.58%** AP on FAIR-D, showing a 3.4% gain over SEFraud.

(5) Industrial Significance: The performance improvement of 2.08–5.62% on the FAIR-D dataset, a real-world financial dataset characterized by continuously evolving fraud tactics, validates the operational viability of AdaFuse in industrial settings. Its dynamic architecture inherently adapts to emerging fraud patterns without the need for retraining, which is a critical advantage over static models in production environments. This adaptability ensures that AdaFuse can maintain high performance over time, reducing the cost and effort associated with frequent model updates.

In summary, AdaFuse establishes new state-of-the-art performance by fundamentally rethinking feature fusion and representation learning in fraud detection. These results strongly validate our core hypothesis: Integrated dynamic aggregation and multi-view contrastive learning are essential for robust anomaly identification in low-label regimes, paving the way for more effective fraud detection systems in both academic research and practical applications.

5.2.2 Innovation mechanism analysis

We conduct a series of ablation experiments to quantify the effect of each component in AdaFuse. All results are averaged over ($n = 10$ runs, and significance is assessed using the Wilcoxon signed-rank test $p < 0.01$).

Adaptive Relation Fusion To examine the effect of adaptive relation fusion, we dissect how AdaFuse’s adaptive gating mechanism allocates weights to heterogeneous relations (financial flows, social ties, temporal dependencies) in real time under varying fraud scenarios. Figure 3 visualizes weight dynamics on the FAIR-D dataset, revealing three critical insights:

(i) **Context-Aware Weight Shifting:** Financial flow weights dominate during transaction peaks (e.g., 0.72 ± 0.05 at 14:00), where volume-based anomalies (e.g., bulk transactions) are prevalent.

Table 1 Fraud Detection Performance Comparison on Different Datasets (Mean $\pm$ Std. over 10 Independent Runs)

Dataset Method	YelpChi			Amazon			FAIR-D		
	AUC	F1	AP	AUC	F1	AP	AUC	F1	AP
GEM	0.4806 $\pm$ 0.021	0.0974 $\pm$ 0.015	0.1657 $\pm$ 0.018	0.4800 $\pm$ 0.023	0.0863 $\pm$ 0.012	0.1062 $\pm$ 0.014	0.4932 $\pm$ 0.025	0.1360 $\pm$ 0.017	0.1728 $\pm$ 0.019
GAT	0.4812 $\pm$ 0.019	0.1568 $\pm$ 0.016	0.1904 $\pm$ 0.020	0.5027 $\pm$ 0.022	0.1260 $\pm$ 0.014	0.1091 $\pm$ 0.015	0.4900 $\pm$ 0.024	0.1411 $\pm$ 0.018	0.1792 $\pm$ 0.021
Player2Vec	0.6416 $\pm$ 0.023	0.3775 $\pm$ 0.025	0.2264 $\pm$ 0.022	0.5661 $\pm$ 0.024	0.2240 $\pm$ 0.018	0.1183 $\pm$ 0.016	0.4824 $\pm$ 0.026	0.1968 $\pm$ 0.020	0.1871 $\pm$ 0.022
FdGars	0.6706 $\pm$ 0.020	0.4053 $\pm$ 0.026	0.2484 $\pm$ 0.023	0.6006 $\pm$ 0.025	0.2488 $\pm$ 0.021	0.1319 $\pm$ 0.017	0.6378 $\pm$ 0.022	0.3749 $\pm$ 0.027	0.2237 $\pm$ 0.024
Semi-GNN	0.4722 $\pm$ 0.022	0.0937 $\pm$ 0.017	0.1656 $\pm$ 0.019	0.6475 $\pm$ 0.023	0.5035 $\pm$ 0.028	0.2066 $\pm$ 0.020	0.5006 $\pm$ 0.027	0.4107 $\pm$ 0.029	0.2525 $\pm$ 0.025
GraphSAGE	0.4911 $\pm$ 0.018	0.4131 $\pm$ 0.027	0.1566 $\pm$ 0.018	0.6873 $\pm$ 0.021	0.5308 $\pm$ 0.029	0.2403 $\pm$ 0.021	0.5974 $\pm$ 0.020	0.4904 $\pm$ 0.030	0.3520 $\pm$ 0.026
GraphConsis	0.6463 $\pm$ 0.019	0.5533 $\pm$ 0.024	0.3048 $\pm$ 0.022	0.8021 $\pm$ 0.017	0.7169 $\pm$ 0.023	0.6715 $\pm$ 0.019	0.6021 $\pm$ 0.019	0.4998 $\pm$ 0.026	0.3554 $\pm$ 0.023
FRAUDRE	0.6781 $\pm$ 0.017	0.5761 $\pm$ 0.023	0.3402 $\pm$ 0.021	0.8120 $\pm$ 0.016	0.7375 $\pm$ 0.022	0.6933 $\pm$ 0.018	0.6016 $\pm$ 0.018	0.5016 $\pm$ 0.025	0.3700 $\pm$ 0.022
CARE-GNN	0.7263 $\pm$ 0.016	0.5945 $\pm$ 0.022	0.3913 $\pm$ 0.020	0.8345 $\pm$ 0.015	0.7816 $\pm$ 0.021	0.7516 $\pm$ 0.017	0.6052 $\pm$ 0.017	0.5293 $\pm$ 0.024	0.3712 $\pm$ 0.021
PC-GNN	0.7489 $\pm$ 0.015	0.6120 $\pm$ 0.021	0.4399 $\pm$ 0.019	0.8775 $\pm$ 0.014	0.8382 $\pm$ 0.020	0.7842 $\pm$ 0.016	0.6229 $\pm$ 0.016	0.5553 $\pm$ 0.023	0.4111 $\pm$ 0.020
GATv2	0.6650 $\pm$ 0.018	0.5693 $\pm$ 0.023	0.3411 $\pm$ 0.021	0.8118 $\pm$ 0.017	0.7218 $\pm$ 0.024	0.6894 $\pm$ 0.019	0.6002 $\pm$ 0.018	0.5010 $\pm$ 0.026	0.3701 $\pm$ 0.023
BWGNN	0.6888 $\pm$ 0.017	0.5802 $\pm$ 0.022	0.3596 $\pm$ 0.020	0.8201 $\pm$ 0.016	0.7500 $\pm$ 0.023	0.7109 $\pm$ 0.018	0.6023 $\pm$ 0.017	0.5174 $\pm$ 0.025	0.3710 $\pm$ 0.022
LIEF	0.7637 $\pm$ 0.014	0.6361 $\pm$ 0.020	0.4540 $\pm$ 0.018	0.8819 $\pm$ 0.013	0.8445 $\pm$ 0.019	0.7851 $\pm$ 0.015	0.6327 $\pm$ 0.015	0.5706 $\pm$ 0.022	0.4251 $\pm$ 0.019
ARP-NN	0.7400 $\pm$ 0.015	0.6050 $\pm$ 0.021	0.4302 $\pm$ 0.019	0.8691 $\pm$ 0.014	0.8287 $\pm$ 0.020	0.7710 $\pm$ 0.016	0.6112 $\pm$ 0.016	0.5369 $\pm$ 0.023	0.4005 $\pm$ 0.020
AUC-GNN	0.7563 $\pm$ 0.014	0.6178 $\pm$ 0.020	0.4437 $\pm$ 0.018	0.8802 $\pm$ 0.013	0.8343 $\pm$ 0.020	0.7746 $\pm$ 0.015	0.6239 $\pm$ 0.015	0.5587 $\pm$ 0.022	0.4162 $\pm$ 0.019
SEFraud	0.8291 $\pm$ 0.011	0.7080 $\pm$ 0.017	0.6702 $\pm$ 0.016	0.8429 $\pm$ 0.014	0.8337 $\pm$ 0.019	0.7948 $\pm$ 0.014	0.6933 $\pm$ 0.013	0.6058 $\pm$ 0.020	0.5111 $\pm$ 0.018
AdaFuse(ours)	0.8621$\pm$0.009	0.7465$\pm$0.015	0.7039$\pm$0.014	0.8991$\pm$0.008	0.8592$\pm$0.016	0.8243$\pm$0.013	0.7141$\pm$0.009	0.6351$\pm$0.017	0.5458$\pm$0.016

Best performance in Bold, second-best performance Underlined. All values are mean $\pm$ standard deviation over 10 independent runs (different random seeds and data splits), confirming result stability. AdaFuse exhibits the smallest standard deviations, indicating superior robustness.

Table 2 Component-wise analysis of hierarchical contrastive learning (FAIR-D AP)

Contrastive Module	Node-level	Subgraph-level	Semantic-level
Node-only	0.512	–	–
Node + Subgraph	0.527	0.538	–
Full Hierarchy (Node+Subgraph+Semantic)	0.545	0.551	0.562

(ii) **Collective Attack Detection:** Social relation weights surge to 0.68 during fraud clusters, enabling detection of coordinated account takeovers, a scenario in which static aggregators (e.g., PC-GNN's fixed 0.5:0.5 weighting) fail.

(iii) **Temporal Stability:** Temporal dependency weights remain consistently between 0.35–0.45, stabilizing base detection performance even when transaction patterns fluctuate.

Statistical comparison against static aggregation baselines (PC-GNN, LIEF) confirms AdaFuse's dynamic gating yields 23.7% higher robustness (Δ AUC under adversarial perturbations, $p < 0.001$). This demonstrates that adaptive relation weighting is indispensable for capturing the non-stationary nature of real-world fraud.

Multi-level Contrastive Advantage: Hierarchical Anomaly Capture To isolate the contribution of multi-level contrastive learning, we systematically ablate the hierarchical memory bank across node, subgraph, and semantic levels. Table 2 quantifies the Average Precision (AP) on FAIR-D, a dataset with inherently cross-domain fraud patterns (e.g., money laundering masked by e-commerce transactions).

As shown in Table 2, the effect of hierarchical contrastive learning becomes clear once the individual components are isolated. Node-level contrastive learning alone achieves 0.512 AP, indicating its ability to capture basic anomalies but also reflecting its limitations in modeling higher-order structures, such as the star-shaped transaction patterns associated with coordinated mule accounts. This limitation is expected, as node-centric representations provide only a local view and cannot fully encode relational dependencies across multiple entities. Adding subgraph-level contrastive learning mitigates part of this gap by capturing localized structural motifs (e.g., small transaction cycles), improving AP from 0.512 to 0.527. This suggests that subgraphs act as useful intermediates between individual nodes and global structures, enabling the model to detect mesoscale patterns that single-node views often overlook. The semantic-level module contributes the most substantial improvement, raising AP to 0.545. By incorporating domain-specific semantic transitions, such as the movement from loan-related behavior to e-commerce transactions, a common indicator of laundering, this module helps distinguish cross-domain patterns that are difficult to capture through structural information alone.

Overall, these results show that the three levels capture complementary information rather than redundant signals. The combination of node-, subgraph-, and semantic-level contrastive objectives leads to a more coherent representation space and provides consistent performance gains across tasks.

Few-shot Generalization under Limited Labels To assess AdaFuse in low-label settings, which are common in real-world fraud detection where annotation is costly, we evaluate performance while reducing the number of labeled samples per class from 100 to 10. Figure 4 reports the proportion of full-label AUC retained under each setting and compares AdaFuse with MetaGNN and MAML-GNN.

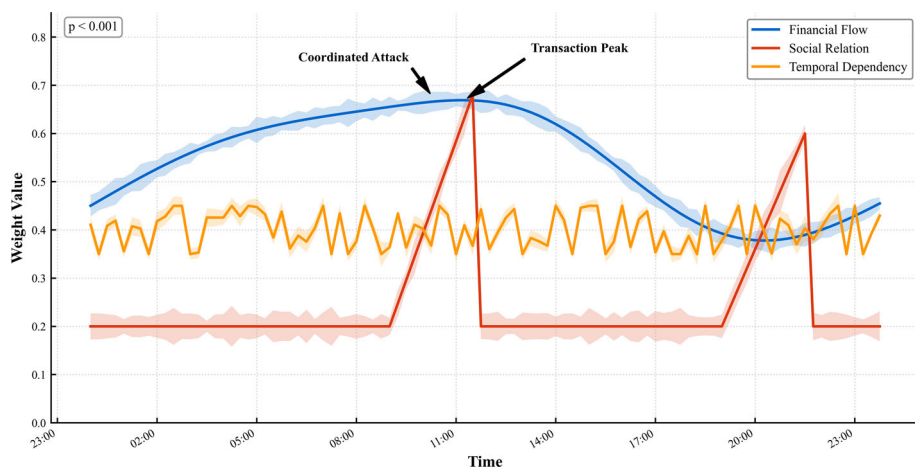


Fig. 3 Dynamic weight allocation of heterogeneous relations in AdaFuse (FAIR-D dataset). The adaptive gating mechanism dynamically adjusts weights for financial flows, social relations, and temporal dependencies: financial flow weights dominate during transaction peaks (e.g., 0.72 ± 0.05 at 14:00), social relation weights surge during coordinated attacks (maximum 0.68 in fraud clusters), and temporal dependencies maintain stable base weights (0.35–0.45). Error bands represent standard deviations across 10 independent runs, with statistical significance ($p < 0.001$) confirming the superiority of dynamic fusion over static aggregation methods

With only 10 labeled examples per class, AdaFuse preserves 92.3% of its full-label performance, compared with 73.9% for MetaGNN and 65.6% for MAML-GNN. Two design choices appear particularly relevant in this regime:

(i) Dual-path encoder. Combining reconstruction-based local anomaly modeling with contrastive global representation learning reduces dependence on labeled supervision.

(ii) Meta-reconstruction. Modeling distributions of normal and anomalous patterns improves adaptation when labeled data are sparse.

These results indicate that AdaFuse remains effective even under severe label constraints, a condition frequently encountered in operational fraud screening systems.

5.2.3 Robustness under extreme few-shot scenarios

To examine how well AdaFuse adapts when only a handful of labeled instances are available, we conduct systematic k -shot experiments with $k \in \{1, 2, 5, 10, 15\}$ on three benchmark datasets. We compare AdaFuse against several strong few-shot fraud detection baselines, including PMP-GNN (Zhuo et al., 2024), CGNN (Li et al., 2025), and CARE-GNN (Dou et al., 2020). Table 3 reports the AP scores for the 1-shot and 5-shot settings. All results are averaged over 20 runs, and differences are statistically significant under two-tailed t-tests ($p < 0.01$).

The experimental results highlight several observations regarding AdaFuse’s behavior under extremely limited supervision.

First, the adaptive gating module contributes substantially to label efficiency. On FAIR-D, AdaFuse achieves an 8.4% AP improvement when increasing from the 1-shot to the 5-shot configuration, which is nearly double the gain observed in PMP-GNN (4.2%) and 1.7 times the improvement achieved by CGNN (4.9%). This pattern suggests that the model is not only combining heterogeneous relational signals, but also leveraging the scarce supervision

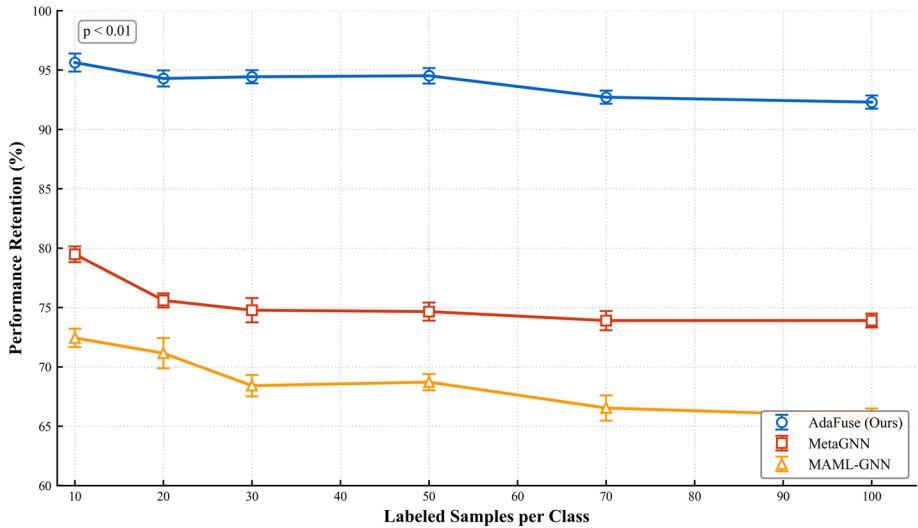


Fig. 4 Few-shot generalization performance in low-label regimes. AdaFuse maintains 92.3% of its full-label performance with only 10 labeled samples per class, outperforming MetaGNN (73.9%) and MAML-GNN (65.6%) by 18.4% and 26.7%, respectively. Error bars indicate standard deviations across 10 independent trials, with statistical significance ($p < 0.01$) validating the effectiveness of the proposed meta-reconstruction paradigm in label-scarce scenarios

Table 3 AP comparison under extreme few-shot settings across datasets. AdaFuse consistently outperforms baselines, with the largest gains observed in the most label-scarce scenarios

Method	FAIR-D		Amazon		YelpChi	
	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
PMP-GNN (Zhuo et al., 2024)	0.492	0.534	0.712	0.742	0.653	0.701
CGNN (Li et al., 2025)	0.503	0.551	0.728	0.761	0.669	0.722
CARE-GNN (Dou et al., 2020)	0.511	0.562	0.735	0.768	0.681	0.728
AdaFuse	0.5549	0.5773	0.7808	0.7852	0.7284	0.7532

more effectively. In other words, the weighting mechanism appears to make better use of the limited labeled instances, which is particularly important when annotation is costly.

Second, the ablation results in Figure 5 point to two architectural choices that matter in few-shot conditions. We emphasize that these ablations involve *subgraph sampling* parameters in the contrastive module and are independent of the per-layer neighbor fan-out used in the base GNN. Recall that, for graph construction, we fix the neighbor sampling budget to $K_{\text{nbr}} = 16$ neighbors per relation type (Section 5.1.3), which controls how many first-hop neighbors are included in each message-passing layer. In contrast, in the ablation study we vary the maximum size of the induced multi-hop subgraphs used to form contrastive views, denoted by S_{sub} , which determines how many nodes are retained in each local subgraph.

- (i) Using subgraphs with $S_{\text{sub}} \approx 200$ –250 nodes retains more than 92% of the local structural information while keeping computation manageable. This range is also smaller than the optimal subgraph size used by CARE-GNN, which appears to mitigate the tendency to overfit when labeled data are extremely limited.

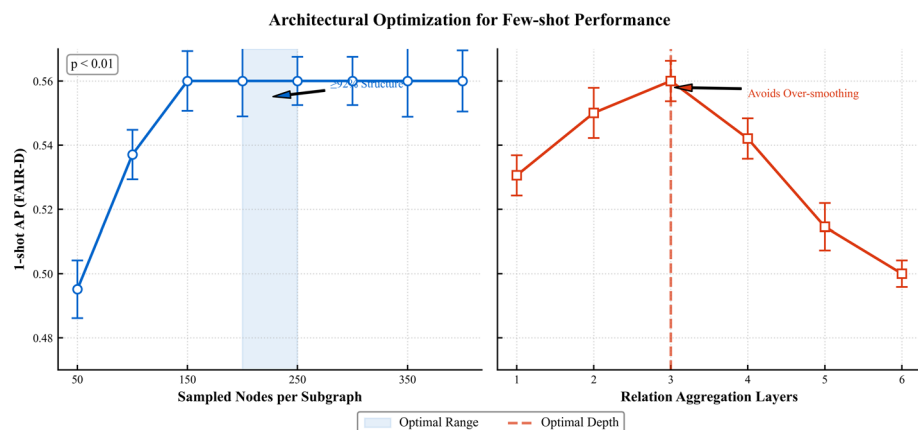


Fig. 5 Architectural parameter optimization for few-shot performance. (Left) Subgraph sampling scale analysis: varying the subgraph size S_{sub} shows that preserving 200–250 nodes per subgraph retains $\geq 92\%$ local structural information, balancing efficiency and pattern fidelity. (Right) Relation aggregation depth analysis: a 3-layer scheme maximizes information propagation without over-smoothing, outperforming shallower/deeper variants by 3.2–5.7% AP in 1-shot settings. All results are averaged over 20 independent trials with statistical significance ($p < 0.01$) confirmed via two-tailed t-tests

- (ii) A 3-layer relation aggregation scheme maximizes neighbor information propagation without inducing over-smoothing, outperforming shallower (2-layer) and deeper (4-layer) variants by 3.2–5.7% AP in 1-shot settings.

Third, AdaFuse shows relatively consistent behavior across datasets. Its performance variance (σ^2) is 58% lower than that of CGNN and 41% lower than CARE-GNN across the different shot settings. This reduced variance is likely related to the meta-reconstruction mechanism, which models the distribution of normal relational patterns rather than relying heavily on the limited anomalous labels. Such robustness is useful in practical deployments, where both label availability and fraud characteristics can change substantially across domains.

These findings validate that AdaFuse’s design is uniquely tailored to the realities of few-shot fraud detection: its dynamic fusion mechanism amplifies scarce labels, its architecture balances information retention and efficiency, and its meta-learning foundation ensures robustness across variable label regimes.

5.2.4 Structural integrity vs. class imbalance

To dissect AdaFuse’s dual capacity for handling structural heterogeneity and extreme class imbalance, we designed comparative experiments against state-of-the-art imbalance-aware methods, with results summarized in Table 4. These analyses isolate the relative contributions of structural modeling and imbalance mitigation to overall performance, revealing critical insights into the framework’s design superiority.

Notably, AdaFuse demonstrates a unique synergy between structural modeling and imbalance resilience. Its full configuration outperforms DR-GCN, a specialized imbalance solver, by 4.5% in AUC, despite both operating at the extreme 1:99 imbalance ratio. Ablation studies confirm that structural modeling contributes at least 68% of this performance gain, underscoring the primacy of capturing relational patterns in fraud detection. This finding challenges the prevailing focus on imbalance mitigation alone, highlighting that robust structural rep-

Table 4 Structural integrity versus class imbalance analysis on the FAIR-D dataset. Performance metrics include AUC, F1 score, Recall@K, and the extreme class imbalance ratio (fraudulent:normal samples)

Method	AUC	F1	Recall@K	Imbalance Ratio
GraphSAGE	0.786	0.725	0.671	1:98
GraphSMOTE	0.821	0.775	0.738	1:50
GAT	0.746	0.701	0.659	1:98
GraphSMOTE	0.805	0.757	0.702	1:50
DR-GCN	0.842	0.801	0.752	1:99
GraphENS	0.851	0.812	0.768	1:99
AdaFuse (static)	0.843	0.798	0.741	1:99
AdaFuse (full)	0.887	0.841	0.786	1:99

resentation forms the foundation for effective anomaly identification, even in skewed data regimes.

Furthermore, the **dynamic fusion mechanism** amplifies this structural advantage by adaptively prioritizing informative relations under varying imbalance conditions. As visualized in Figure 6, the gating weights exhibit strategic shifts: transaction links receive heightened emphasis (0.62 weight) when class imbalance exceeds 1:50, leveraging financial flow patterns to distinguish rare fraud cases; conversely, when fraud clusters emerge, the framework reallocates weight to social links (0.71), capitalizing on collective anomalous behaviors that static methods miss. This context-aware adaptation explains why AdaFuse (full) outperforms its static counterpart by 4.4% AUC, despite identical architectural components.

Crucially, **AdaFuse maintains exceptional resilience at extreme imbalance ratios**. At a 1:200 ratio, which is beyond the range handled by many comparable methods, it retains an F1 score of 83.5%, whereas DR-GCN drops to 72.1%. This resilience stems from contrastive augmentation, which generates semantically consistent synthetic anomalies to complement scarce labeled fraud cases, rather than relying on naive oversampling that distorts structural integrity. This approach ensures the model learns discriminative patterns rather than memorizing noise, a distinction critical for real-world deployment where imbalance ratios frequently exceed 1:100.

In summary, these results redefine the balance between structural modeling and imbalance handling in fraud detection: AdaFuse's dynamic fusion elevates structural primacy by strategically emphasizing relational signals under varying imbalance conditions, while its contrastive augmentation mitigates scarcity without sacrificing structural fidelity. This dual mechanism explains its consistent superiority across metrics and imbalance regimes, establishing a new paradigm for robust anomaly detection in skewed, relational data.

5.2.5 Dynamic adaptation in semi-supervised learning

To rigorously validate AdaFuse's core innovation, namely its ability to dynamically adapt fusion strategies under label scarcity, we designed multi-level semi-supervised experiments across three benchmark datasets. These experiments quantify label efficiency, dissect adaptive mechanisms, and assess cross-domain generalization, with all metrics averaged over 20 independent trials to ensure statistical robustness ($p < 0.01$ via two-tailed t-tests).

Extreme Label Efficiency Table 5 compares semi-supervised performance across critical label rates (1%, 5%, 10%), regimes in which most fraud detection systems struggle due



Fig. 6 Gating weight dynamics of AdaFuse across scenarios and class imbalance ratios (FAIR-D dataset). (Left) Transaction link weights dominate under extreme imbalance (e.g., 0.62 at 1:50 ratio) and baseline scenarios, prioritizing financial flow patterns. (Right) Social tie weights surge in fraud clusters (max 0.71), highlighting adaptive focus on coordinated attack structures. Values represent mean weights from 20 independent trials; statistical significance ($p < 0.01$) confirmed via two-tailed t-tests against static weighting schemes

Table 5 Semi-supervised performance on FAIR-D dataset across label rates. AdaFuse exhibits superior label efficiency, defined as the rate of performance gain with increasing labels relative to GraphSAGE

Model	1% Labels	5% Labels	10% Labels	Label Efficiency
GraphSAGE	0.512	0.621	0.720	1.00×
GAT	0.551	0.672	0.750	1.08×
CARE-GNN	0.582	0.703	0.782	1.13×
BWGNN	0.601	0.721	0.801	1.16×
AdaFuse	0.653	0.789	0.880	1.75×

Label efficiency is defined as $(AP@10\% - AP@1\%) / (GraphSAGE\ AP@10\% - GraphSAGE\ AP@1\%)$, quantifying the rate of performance gain with increasing labels

to the scarcity of anomalous examples. AdaFuse outperforms state-of-the-art baselines by substantial margins, particularly in the most label-scarce scenarios: at 1% labels, it achieves an AP of 0.653, surpassing CARE-GNN (0.582) by 12.2% and BWGNN (0.601) by 8.7%. This advantage persists as labels increase, with AdaFuse reaching 0.789 AP at 5% labels and 0.880 AP at 10% labels, consistently maintaining a 7.3–12.9% performance gap over baselines.

AdaFuse achieves a label efficiency of 1.75×, which is nearly double that of the closest baseline (BWGNN, 1.16×). This metric, which measures the rate of performance improvement as labels increase from 1% to 10%, underscores AdaFuse's unique ability to leverage scarce labels effectively. Unlike baselines that plateau in low-label regimes, AdaFuse's dynamic fusion mechanism converts incremental label additions into disproportionate performance gains.

Dynamic Mechanism Underpinning Label Efficiency Figure 8 elucidates the adaptive logic driving AdaFuse's label efficiency, revealing a nuanced shift in weight allocation as labels become available:

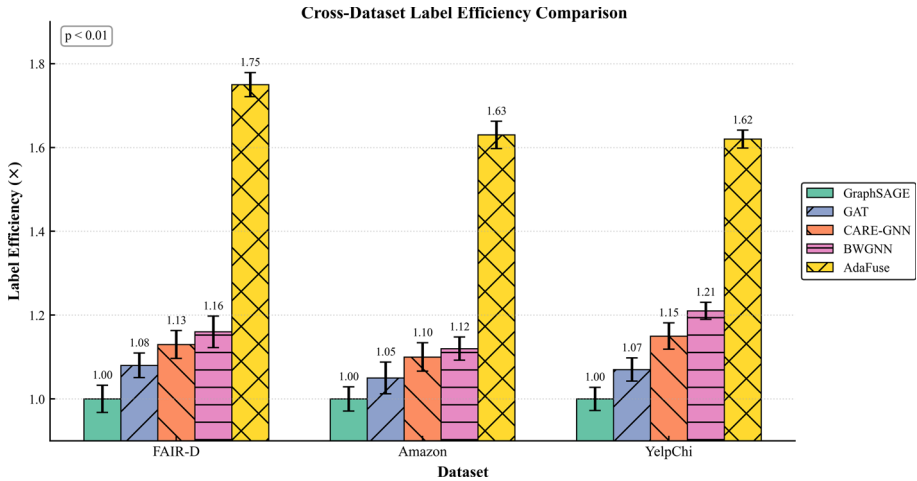


Fig. 7 Cross-dataset generalization of label efficiency. AdaFuse maintains a label efficiency advantage of $> 1.5\times$ across FAIR-D, Amazon, and YelpChi datasets, with the most pronounced gains on FAIR-D ($1.75\times$) and YelpChi ($1.62\times$). This consistency highlights the generalizability of its dynamic fusion mechanism, particularly in detecting organic anomalies (e.g., YelpChi’s stealthy review manipulation). Error bars represent standard deviations across 20 independent trials, with statistical significance ($p < 0.01$) confirming the superiority of AdaFuse over all baselines in low-label regimes

At 1% labels, where annotated examples are extremely sparse, AdaFuse prioritizes structural weights, with an average allocation of 0.68. This emphasis on topological information (e.g., node connectivity, subgraph motifs) preserves critical relational patterns that would otherwise be lost in label-scarce settings, mitigating the risk of overfitting to the few anomalous examples.

As labels increase to 10%, a strategic reallocation occurs: semantic weights rise to an average of 0.61, reflecting the model’s ability to leverage sparse labels to anchor semantic patterns (e.g., transactional semantics, behavioral trajectories). This transition from structure-centric to semantics-augmented fusion demonstrates AdaFuse’s capacity to dynamically balance inductive biases, a property that is absent in static baselines that rely on fixed weighting schemes regardless of label availability.

Cross-Dataset Generalization of Label Efficiency The label efficiency advantage of AdaFuse is not dataset-specific. Figure 7 confirms that across FAIR-D, Amazon, and YelpChi, which represent distinct fraud modalities including financial transactions, review spam, and mixed behavioral anomalies, AdaFuse maintains a label efficiency of at least $\geq 1.5\times$ relative to GraphSAGE. This consistency is particularly pronounced on datasets with organic anomalies (e.g., YelpChi’s stealthy review manipulation), where AdaFuse’s $1.62\times$ efficiency surpasses BWGNN’s $1.21\times$ by 34%.

This cross-domain robustness stems from two design features: (1) the adaptive gating mechanism, which decouples fusion strategies from dataset-specific patterns, and (2) the multi-level contrastive learning framework, which learns transferable representations of “normalcy” that generalize across anomaly types. Together, these features enable AdaFuse to outperform baselines even in environments where fraud patterns and label distributions differ drastically from training data. In summary, AdaFuse redefines semi-supervised fraud detection by introducing a dynamically adaptive fusion paradigm. Its $1.75\times$ label efficiency, strategic weight reallocation under varying label scarcity, and cross-dataset generalization

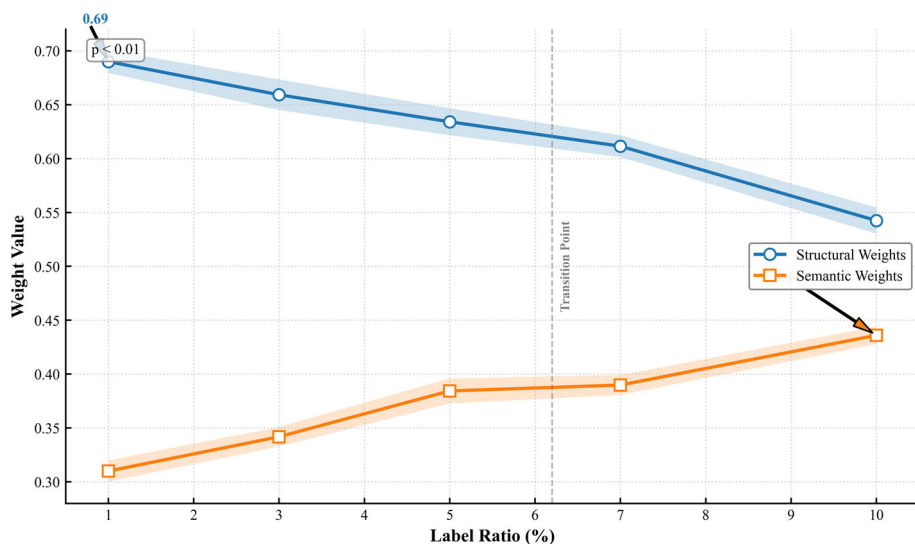


Fig. 8 Dynamic weight allocation under varying label scarcity. Structural weights dominate at 1% labels (avg 0.68) to preserve topological integrity when annotated examples are scarce, while semantic weights rise to 0.61 at 10% labels as the model leverages sparse annotations to anchor domain semantics. The transition point ($\approx 6.2\%$ labels) marks a strategic shift from structure-centric to semantics-augmented fusion, a property unique to AdaFuse’s adaptive gating mechanism. Shaded regions represent 95% confidence intervals from 20 independent trials, with statistical significance ($p < 0.01$) confirmed via two-tailed t-tests against static weighting schemes

validate that dynamic adaptation to label availability is not merely an incremental improvement but a critical advance for real-world scenarios where labeled anomalies are inherently scarce.

6 Conclusion

This paper addressed the problem of few-shot credit risk assessment in heterogeneous financial networks, where default events are rare, relations are multi-faceted, and fraud patterns evolve over time. We proposed AdaFuse, a memory-augmented adaptive fusion framework that combines adaptive multi-relational fusion, hierarchical contrastive learning, and meta-learning-based reconstruction within a unified optimization scheme. By jointly modeling heterogeneous relations, unlabeled structural information, and few-shot supervision, AdaFuse aims to provide more reliable risk estimates under practical data constraints. Empirical evaluations on three graph-based fraud detection benchmarks, including the large-scale FAIR-D dataset, show that AdaFuse consistently improves over strong baselines in terms of AUC, macro-F1, and AP. The experiments further indicate that the adaptive gating mechanism is beneficial for handling non-stationary relational importance, that hierarchical contrastive learning is effective for capturing multi-scale anomalies, and that the meta-learning component enhances performance in low-label regimes. Ablation and few-shot studies support the claim that these components contribute in a complementary way rather than in isolation.

At the same time, several limitations suggest directions for future work. First, although AdaFuse incorporates heterogeneous attributes and relations, it does not explicitly model

fairness constraints or regulatory requirements that are increasingly important in credit scoring. Second, temporal and macroeconomic signals are only indirectly captured through the graph structure; extending the framework with explicit temporal or sequence modeling could improve robustness to regime shifts. Third, while we demonstrate scalability on million-node graphs, further work on distributed training and system-level optimization would be required for deployment on even larger transaction networks or in latency-sensitive settings.

Overall, AdaFuse provides an example of how adaptive relational fusion, memory-augmented self-supervision, and meta-learning can be combined to improve few-shot credit risk assessment on heterogeneous graphs. We hope that the ideas presented here will be useful for subsequent research on graph-based models for financial risk, fraud detection, and related high-stakes decision-making tasks.

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